

### Code:

```
#include <iostream>
#include <vector>
using namespace std;
int main() {
    int n, m;
    cout << "Enter number of processes: ";
    cin >> n;
    cout << "Enter number of resource types: ";
    cin >> m;
    vector<vector<int>> alloc(n, vector<int>(m));
    vector<vector<int>> maxm(n, vector<int>(m));
    vector<vector<int>> need(n, vector<int>(m));
    vector<int> avail(m);
    cout << "\nEnter Allocation Matrix:\n";
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            cin >> alloc[i][j];
    cout << "\nEnter Maximum Matrix:\n";
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            cin >> maxm[i][j];
    cout << "\nEnter Available Resources:\n";
    for (int j = 0; j < m; j++)
        cin >> avail[j];
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            need[i][j] = maxm[i][j] - alloc[i][j];
    vector<bool> finish(n, false);
    vector<int> safeSeq;
    vector<int> work = avail;
    for (int count = 0; count < n; count++) {
        bool found = false;
        for (int i = 0; i < n; i++) {
            if (!finish[i]) {
                bool possible = true;
                for (int j = 0; j < m; j++) {
                    if (need[i][j] > work[j]) {
                        possible = false;
                        break;
                    }
                }
            }
            if (possible) {
                for (int j = 0; j < m; j++)
                    work[j] += alloc[i][j];
                safeSeq.push_back(i);
            }
        }
    }
}
```

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        finish[i] = true;
        found = true;
    }
}
}
if (!found)
    break; }
if (safeSeq.size() == n) {
    cout << "\nSystem is in a SAFE state.\n";
    cout << "Safe Sequence: ";
    for (int i = 0; i < n; i++) {
        cout << "P" << safeSeq[i];
        if (i != n - 1)
            cout << " -> "; }
    cout << endl;
} else {
    cout << "\nSystem is NOT in a SAFE state.\n";
}
return 0;
}

```

## Output:

```

● Enter number of processes: 5
  Enter number of resource types: 3

  Enter Allocation Matrix:
  0 1 0
  2 0 0
  3 0 2
  2 1 1
  0 0 2

  Enter Maximum Matrix:
  7 5 3
  3 2 2
  9 0 2
  2 2 2
  4 3 3

  Enter Available Resources:
  3 3 2

  System is in a SAFE state.
  Safe Sequence: P1 -> P3 -> P4 -> P0 -> P2

```